



*Your turning point in the AI era.*

ENGINEERING OVERVIEW

# Retail Support RAG Service

*A grounded support assistant that answers retail ops questions from the company's own catalogs, SOPs and ticket history — with sources attached.*

Python

Claude

Pinecone

~55%

fewer escalations to senior engineers

3

knowledge sources ingested

100%

of answers cite a source document

Auto

escalation with context attached

## THE PROBLEM

# Routine answers are buried, so everything escalates

Frontline retail staff escalate routine questions to senior engineers because the answers are buried across catalogs, SOPs and old support tickets.

## Product Catalogs

Specs, pricing and availability in PDFs no one searches

## SOPs

Process docs that are never consulted at point of need

## Ticket History

Resolved tickets with answers already in them

## Internal Wikis

Knowledge that exists but is never surfaced

## Hours

of senior engineering time  
spent on questions with known answers

### No single

place to search

### Answers

without sources

### Escalation

as the default path

*"Half of what gets escalated to me is a question the answer already exists for. It's just buried somewhere no one knows to look."*

— Composite senior engineer feedback

## HOW IT WORKS

# Four steps, from raw documents to a grounded, sourced answer

01

## Ingest

Product catalogs, SOPs and resolved tickets are chunked, embedded and stored in a Pinecone vector store.

Catalogs

SOPs

Tickets

Chunking

02

## Retrieve

Each incoming query is embedded and the most relevant chunks are retrieved by cosine similarity.

Embedding

Cosine search

Top-k chunks

03

## Answer

Claude answers using only the retrieved context and attaches the source document name to every response.

Grounded gen

Citation attach

Confidence score

04

## Escalate

Low-confidence or out-of-scope queries are routed to a human with the gathered context already attached.

Human handoff

Context bundle

Ticket creation

## KEY FEATURES

# Four capabilities that earn their place

*No hallucinations, no blind escalations. Every feature serves the frontline query.*

1

## Vector Store over Three Knowledge Sources

Product catalogs, SOPs and resolved tickets are chunked, embedded and stored in Pinecone — one search across all three.

Catalogs

SOPs

Tickets

Pinecone

2

## Source Citation on Every Answer

Claude cites the source document name in every response — frontline staff can verify the answer and trust it.

Doc citation

Verifiable

3

## Confidence-Gated Escalation

Low-confidence responses trigger an automatic escalation to a human — with the retrieved context already bundled in.

Auto escalate

Context bundled

4

## FastAPI Endpoint, Drops into Existing Tools

A single FastAPI endpoint integrates into any existing support tool — ticketing system, chat platform or internal portal.

FastAPI

Plug-and-play

## ARCHITECTURE

# Python/FastAPI service with Pinecone for retrieval and Claude API for grounded generation, versioned in Git

## INGESTION (Python pipeline)

Product catalogs

SOPs

Resolved tickets

Git versioned

## RETRIEVAL (Pinecone)

Query embedding

Cosine search

Top-k rerank

Context bundle

## GENERATION (Claude API)

Grounded prompt

Citation inject

Confidence score

Escalation gate

## DELIVERY

FastAPI response · Cited answer · Source doc · Confidence score · Human escalation path

## RESULTS &amp; IMPACT

# Three numbers that justify the build

~55%

fewer escalations to senior engineers

Consistent

sourced answers for frontline staff

Senior

engineering time freed for real work

## What this unlocks

- Frontline staff get answers in seconds, not via a ticket to engineering
- Every answer is traceable to a source — staff trust the response
- Low-confidence queries still go to a human — with context already bundled
- Knowledge stays current as catalogs and SOPs are re-ingested
- Escalation rate drops as the vector store covers more known queries

*“The questions that reach me now are genuinely hard. The ones with known answers never make it through anymore.”*

— Composite senior engineer feedback



*Your turning point in the AI era.*

**Every question answered.  
Every answer sourced.  
Right person every time.**

*The Retail Support RAG Service grounds every answer in the company's own knowledge — with sources attached and a confidence gate before any human handoff.*

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**Thank you.**